TITE-BOLD with prediction-guided intervention

Pending untreated outcomes are replaced by observations; prevented-toxicity counterfactuals remain latent.

ENROLLMENT — **PARTIAL FOLLOW-UP** — **OUTCOME RESOLUTION**

A No preventive intervention

Prediction + pending outcome

$p_j \mid h \sim \mathrm{Beta}(A_h, B_h)$

$q(t) = 1 - t/W$

$z_i = 1$

$\mathrm{Beta}(A_h + 1, B_h), \;\; r_{h1}(t)$

Observed DLT: $Y_i = 1$

$(A_h, B_h) \longmapsto (A_h + 1, B_h)$

$z_i = 0$

$\mathrm{Beta}(A_h, B_h + 1), \;\; r_{h0}(t)$

No DLT through W**:** $Y_i = 0$

$(A_h, B_h) \longmapsto (A_h, B_h + 1)$

Replace this patient's pending label/scores; rebuild from observed history.

B Preventive intervention after a positive prediction

$L_i = +$**; intervene**

$q = 1$

Untreated status is unknown

$z_i = 1$

$\mathrm{Beta}(A_h + 1, B_h), \;\; r_{h1}$

Observed intervention failure

$Y_i = 1 : \quad (A_h, B_h) \mapsto (A_h + 1, B_h)$

$z_i = 0$

$\mathrm{Beta}(A_h, B_h + 1), \;\; r_{h0}$

No observed DLT after intervention

$Y_i = 0 / \Rightarrow \;\; z_i = 0$

Both untreated-outcome branches remain; $q = 1$.

Observed outcomes replace predictions only when untreated status becomes known. After prevention, the unobserved counterfactual does not resolve.

Current combined protocol: intervened patients finish follow-up before the next dose decision; other outcomes may remain pending.

Shared weighted Beta-mixture update

$$A_h = a_j + y_j + k_h, \quad B_h = b_j + n_j^{\mathrm{obs}} - y_j + m_h - k_h; \qquad p_j \mid h \sim \mathrm{Beta}(A_h, B_h), \quad w_h \geq 0, \quad \sum_h w_h = 1$$

Untreated DLT child: $z = 1$

$$\mathrm{Beta}(A_h + 1, B_h), \quad \theta_{h1} = \frac{A_h + 1}{A_h + B_h + 1}$$

$$s_{h1} = \frac{\ell_1 q\, \theta_{h1}}{\ell_1 q\, \theta_{h1} + \ell_0 (1 - \theta_{h1})}, \qquad r_{h1} = w_h s_{h1}$$

Untreated non-DLT child: $z = 0$

$$\mathrm{Beta}(A_h, B_h + 1), \quad \theta_{h0} = \frac{A_h}{A_h + B_h + 1}$$

$$s_{h0} = \frac{\ell_0 (1 - \theta_{h0})}{\ell_1 q\, \theta_{h0} + \ell_0 (1 - \theta_{h0})}, \qquad r_{h0} = w_h s_{h0}$$

$$\widetilde{w}_{hz} = \frac{r_{hz}}{\sum_{h'} \sum_{z'=0}^{1} r_{h'z'}}, \qquad f_{\mathrm{new}}(p_j) = \sum_h \sum_{z=0}^{1} \widetilde{w}_{hz}\, \mathrm{Beta}(p_j; A_h + z, B_h + 1 - z)$$

$L = +: (\ell_1, \ell_0) = (s, 1 - c); \quad L = -: (1 - s, c); \quad$ no label $: (1, 1). \qquad a_j = \mu_j \nu_j, \quad b_j = (1 - \mu_j)\nu_j.$

s: sensitivity; c: specificity; W: DLT window; w_h: parent weight; m_h: prior latent patients; k_h: their hypothesized DLT count.